UNIVERSITY OF SOUTH CAROLINA
College of Engineering and Computing

EMCH 260
SOLID MECHANICS

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Torsion

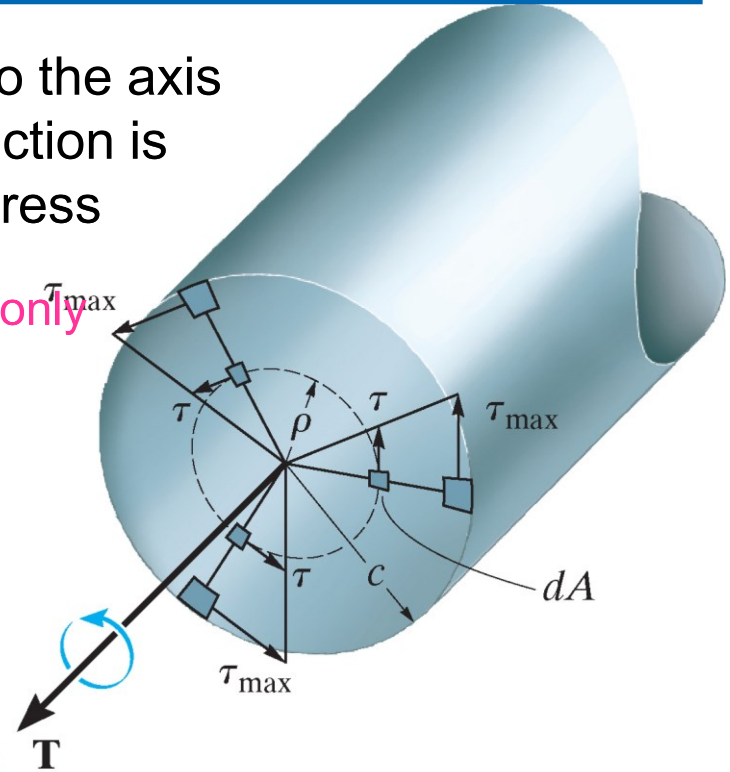
5

Chapter 5.2

- ✓ Determine the angle of twist
- ✓ Analyze the angle of twist in single and multiple torques, and possibly with different cross-sectional area

PREVIOUS ON TORSION FORMULA

- For any cross section that is perpendicular to the axis of the shaft, the torque in the entire cross section is contributed by the internal resultant shear stress



Internal resultant torque   For circular cross-section only

$$\boxed{T \equiv \int_A \rho(\tau dA)} \boxed{= \int_A \rho\left(\frac{\rho}{c}\right)\tau_{max} dA}$$

→ $$\tau_{max} = \frac{Tc}{J}$$

or $$\tau = \frac{T\rho}{J}$$

Shear stress distribution: τ increases linearly with radial distance

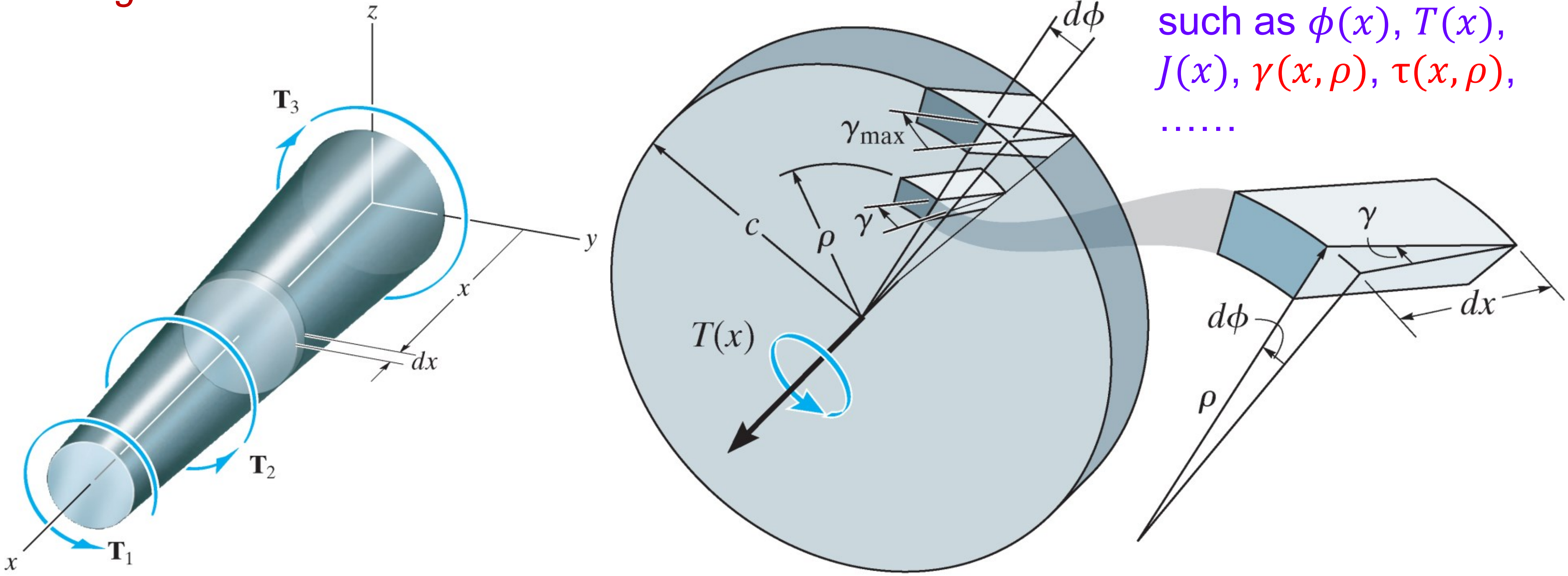
Note that here both the internal torque and polar moment of inertia are constant along the axis direction

J: ***polar moment of inertia*** *(pure geometry property of cross-section)*

ANGLE OF TWIST (GENERAL CASE)

<u>When</u> both the internal torque and polar moment of inertia are not constant along the axis direction…

Most of properties will change along the axis direction, such as $\phi(x)$, $T(x)$, $J(x)$, $\gamma(x,\rho)$, $\tau(x,\rho)$, ……



Geometry relationship still holds

- For any cross section located at x, $d\phi = \gamma \dfrac{dx}{\rho}$

ANGLE OF TWIST, CONT'D

- *Using Hooke's law*: $\tau(\rho, x) = G(x) \cdot \gamma(\rho, x)$

$$\rightarrow \quad d\phi = \frac{\tau(\rho, x)}{G(x)} \frac{dx}{\rho}$$

$$\tau(\rho, x) = \frac{T(x)\rho}{J(x)}$$

(for given position of cross section, recall the torsion formula)

$$\rightarrow \quad d\phi = \frac{T(x)}{J(x)G(x)} dx$$

Integral form (twist angle of one end with respect to the other end):

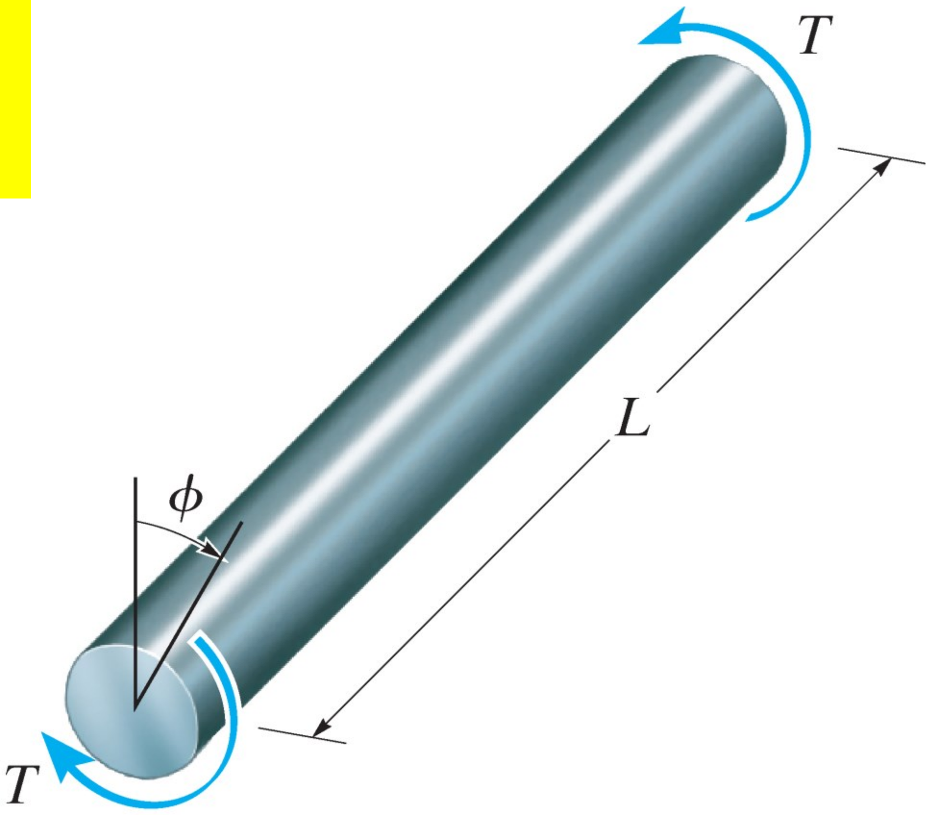
$$\boxed{\phi \equiv \int_0^L \frac{T(x)}{J(x)G(x)} dx}$$

Note that ϕ only depends on axial position x, does not depend on radial position ρ. ***WHY?***

CONSTANT TORQUE AND CROSS-SECTIONAL AREA

- *For special case of constant torque and cross-sectional area along the axis direction,* $T(x)$, $J(x)$, and $G(x)$ are all constant.

$$\phi = \frac{T}{JG}\int_0^L dx = \frac{TL}{JG}$$

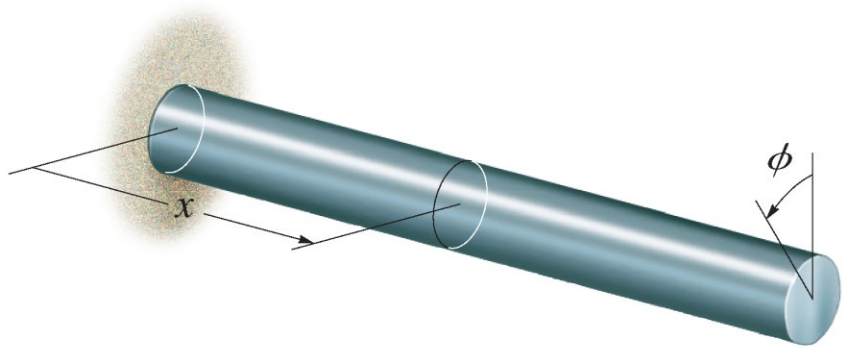


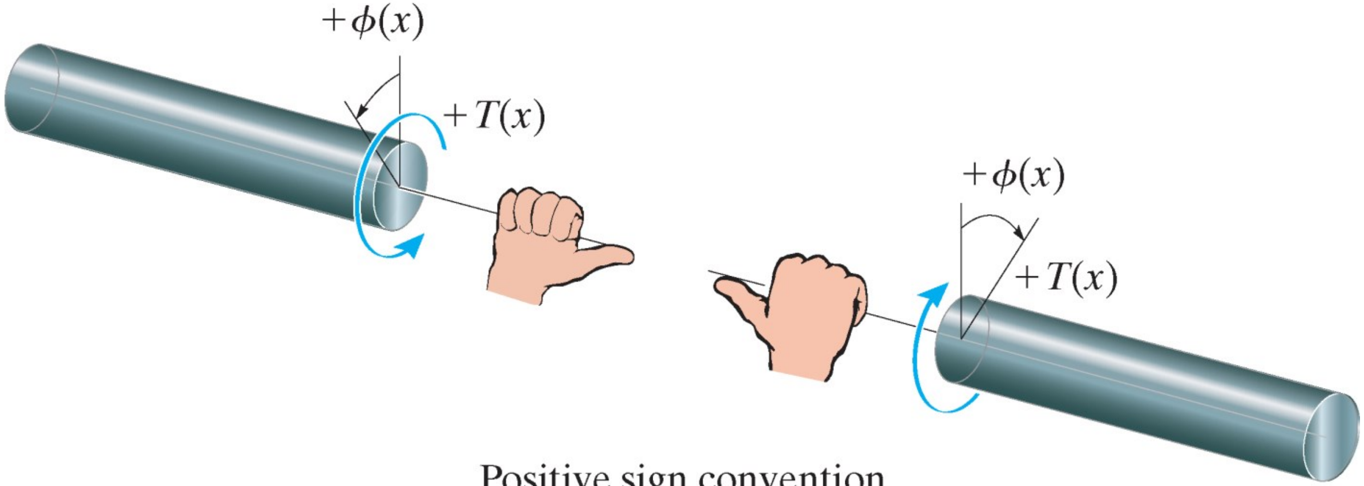
MULTIPLE TORQUES

- If the shaft is subjected to *several different torques*, or the cross-sectional area or shear modulus changes *abruptly* from one region of the shaft to the next, then each region can be considered as the special case of constant torque and cross-sectional area. The total angle of twist of one end of the shaft with respect to the other is equal to the algebraic addition of the angles of twist of each segment or region.

$$\phi = \sum \frac{TL}{JG}$$

SIGN CONVENTION





Positive sign convention for T and ϕ.

Right-hand-rule

- Both the torque and twist angle will be *positive*, provided the *thumb* is directed *outward* from the shaft while the fingers curl in the direction of the torque.

PROCEDURE FOR ANALYSIS

Step 1: Internal Torque

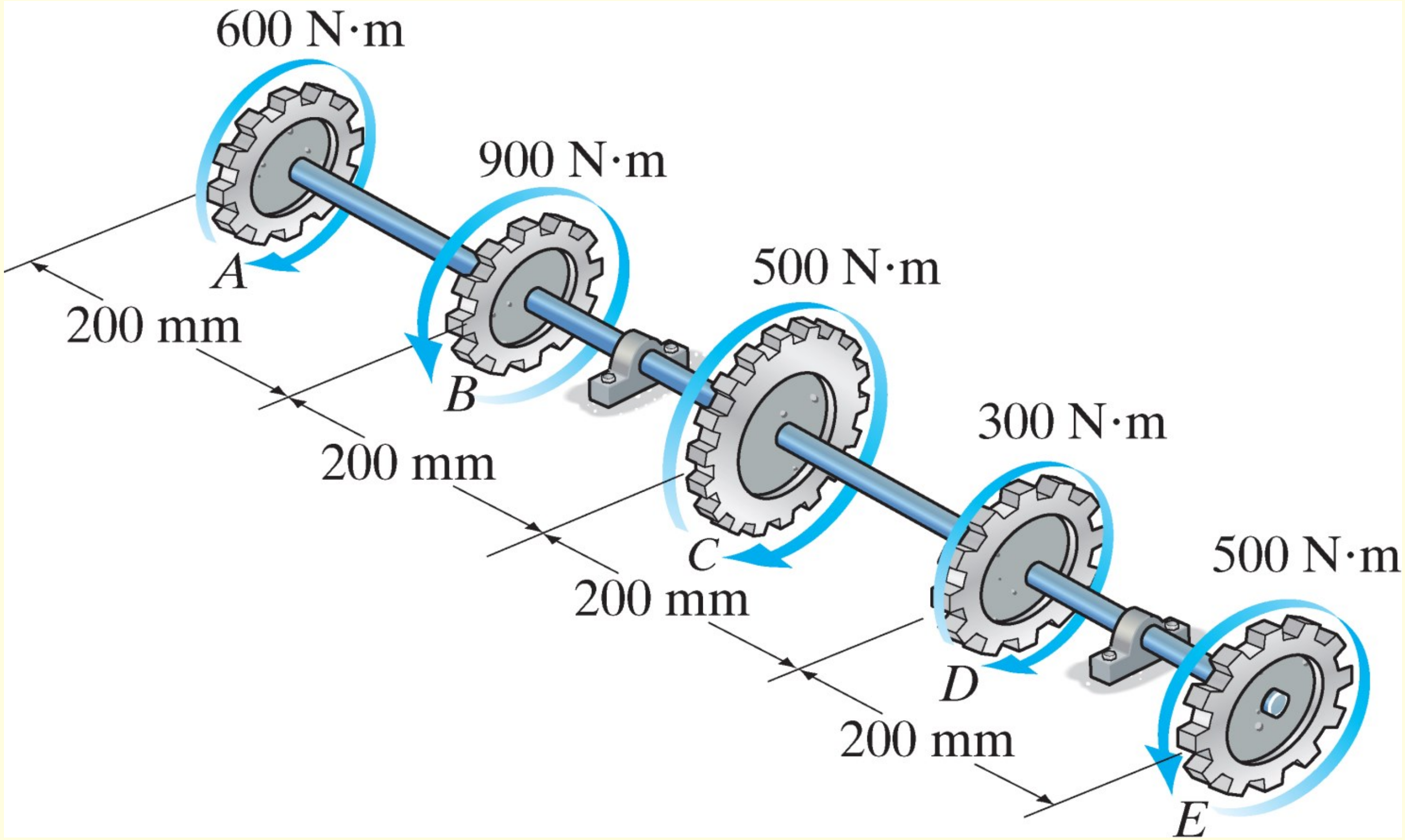
- Use the section method and equation of moment equilibrium, applied along the shaft's axis, to determine the internal torque.
- Note that sometimes the torque varies along the shaft's length, i.e. $T(x)$.

Step 2: Angle of Twist

- When the circular cross-sectional area of the shaft varies along the axis, the polar moment of inertia also depends on its position *x* along the axis, i.e. $J(x)$.
- Calculate the angle of twist using either the general formula $\phi = \int_0^L \frac{T(x)}{J(x)G(x)} dx$ or the formula under special circumstances $\phi = \frac{TL}{JG}$ or $\phi = \sum \frac{TL}{JG}$.
- Make sure to use a consistent sign convention for the shaft or its segments.

IN-CLASS EXAMPLE 1

A series of gears are mounted on the 40-mm-diameter steel shaft with shear modulus of elasticity $G = 75\ \mathrm{GPa}$. Determine the angle of twist of gear *E* relative to gear *A*.

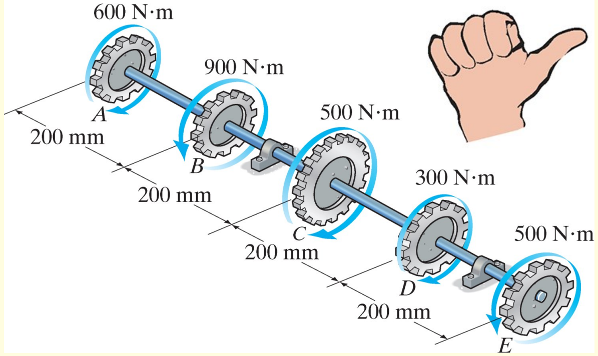


IN-CLASS EXAMPLE 1 (cont'd)

Solutions

Step 1: Internal Torque

Using the method of sections, the internal resultant torques are found in each segment as shown below. By the right-hand rule, with *positive* torques directed *away from the sectioned end of the shaft*, we have



$$T_{AB} = +600\ \mathrm{N \cdot m}$$

$$T_{BC} = -300\ \mathrm{N \cdot m}$$

$$T_{CD} = +200\ \mathrm{N \cdot m}$$

$$T_{DE} = +500\ \mathrm{N \cdot m}$$

For details how to determine the internal torques and in particular their **sign**, refer to Example 5.5, textbook, p. 211

NOTE:
The sign does not depend on which part you use on the two sides of the section

Step 2: Angle of Twist

The polar moment of inertia for the shaft is

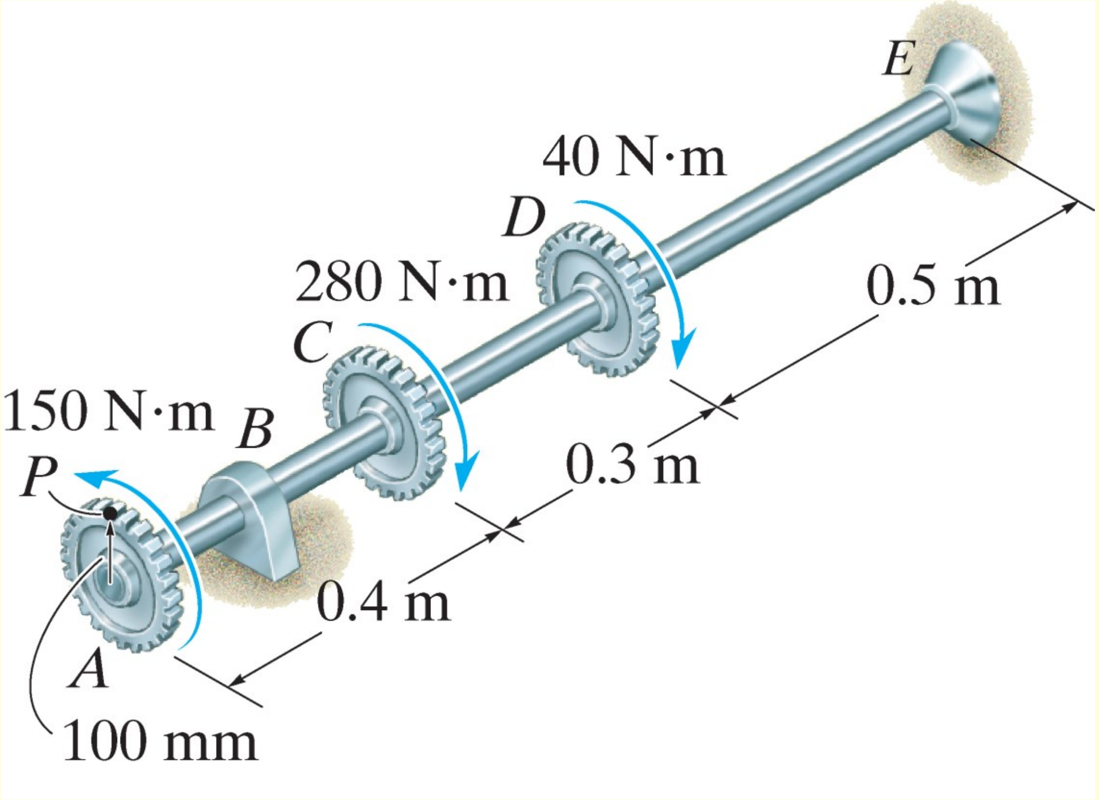
$$J = \frac{\pi}{2} \times 0.02^4 = 2.5133 \times 10^{-7}\ \mathrm{m}^4$$

The angle of twist of gear *E* relative to gear *A* involves the rotation of all four segments

$$\phi_{A/E} = \sum \frac{TL}{JG} = \frac{0.2}{2.5133 \times 10^{-7} \times 75 \times 10^9} \times (600 - 300 + 200 + 500) = 0.01061\ \mathrm{rad}$$ **(Ans.)**

IN-CLASS EXAMPLE 2

The gears attached to the fixed-end steel shaft are subjected to the torques. If the shaft has a diameter of 14 mm, determine the displacement of the tooth *P* on gear *A*. Shear modulus of elasticity $G = 80\ \text{GPa}$.



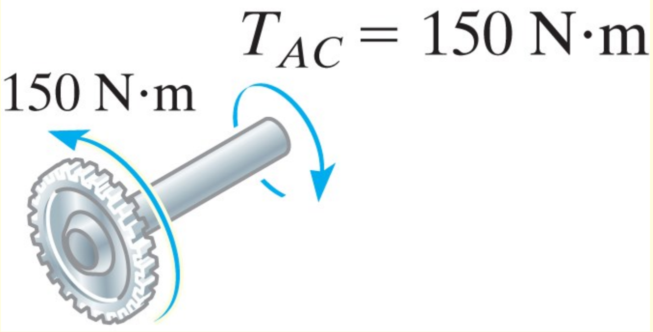
Note the slight difference between Example 1 and 2!

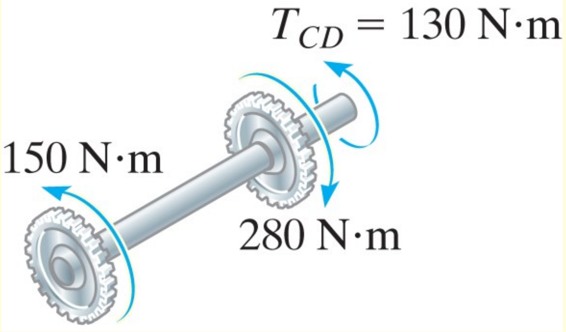
IN-CLASS EXAMPLE 2 (cont'd)

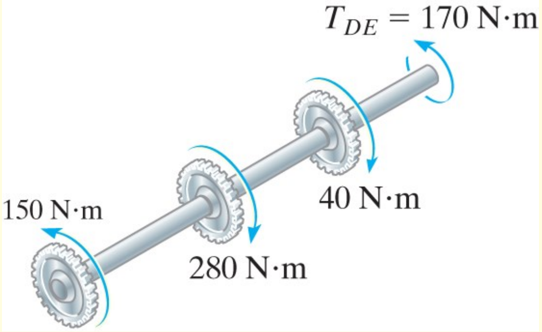
Solutions

Step 1: Internal Torque

Using the method of sections and the right-hand rule, the internal resultant torques in each segment AC, CD, and DE can be determined as







$$T_{AC} = +150\ \mathrm{N \cdot m} \qquad T_{CD} = -130\ \mathrm{N \cdot m} \qquad T_{DE} = -170\ \mathrm{N \cdot m}$$

IN-CLASS EXAMPLE 2 (cont'd)

Solutions

Step 2: Angle of Twist

The polar moment of inertia for the shaft is

$$J = \frac{\pi}{2} \times 0.007^4 = 3.7715 \times 10^{-9}\ \mathrm{m}^4$$

The angle of twist of gear *A* relative to end *E* involves the rotation of all three segments

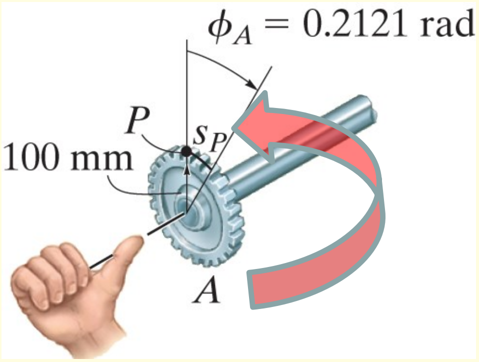
$$\phi_{A/E} = \sum \frac{TL}{JG} = \frac{1}{3.7715 \times 10^{-9} \times 80 \times 10^9} \times (150 \times 0.4 - 130 \times 0.3 - 170 \times 0.5)$$

$$= -0.2121\ \mathrm{rad}$$

Since the end *E* is fixed, this angle of twist is exactly the absolute rotation of gear *A*. Therefore, the displacement of tooth *P* on gear *A* is

$$s_P = \phi_{A/E} r = -0.2121 \times 0.1$$

$$= -0.02121\ \mathrm{m} = -21.21\ \mathrm{mm}$$ **(Ans.)**



The negative sign indicates the gear *A* rotates in the opposite direction of the torque applied to gear *A*.

RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #4 (go to Blackboard under the "Assignments" folder, Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021)**

- **Optional Extra Exercise 2 on Blackboard, problem #3 (go to Blackboard under the "Extra Exercises" folder, Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021)**

OUT-OF-CLASS READING

- **Textbook, pp. 206 – 214**
  - **In particular, Examples 5.5 – 5.6 (pp. 211 – 212)**